

1. (5.20)

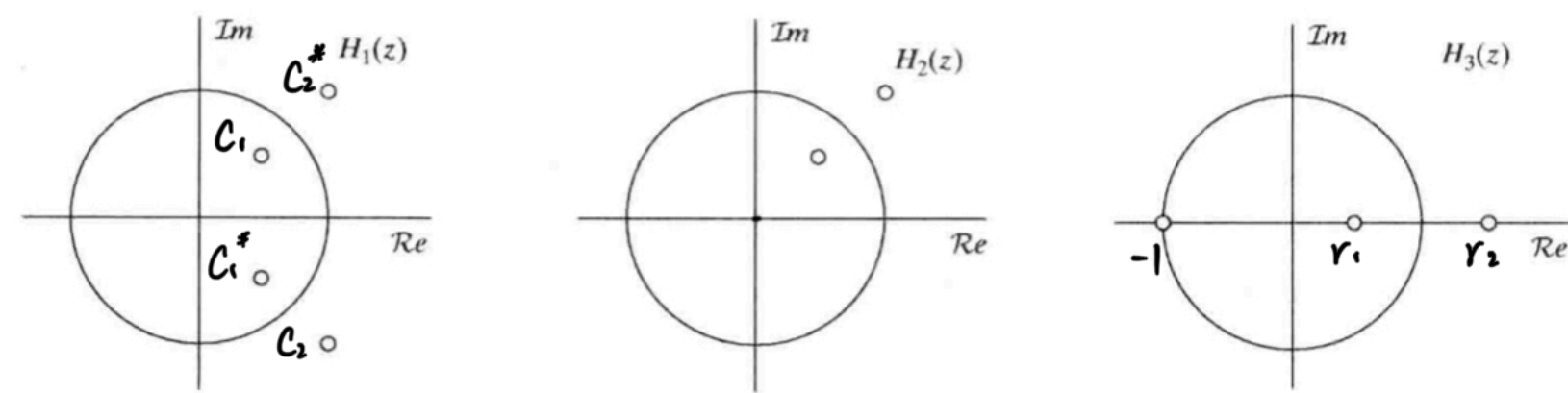


图 P5.20

(a)  $H_1(z) = (1 - c_1 z^{-1})(1 - c_2 z^{-1})(1 - c_1^* z^{-1})(1 - c_2^* z^{-1})$  [能组成实系数线性差分方程]  
 $= [1 - (c_1 + c_2)z^{-1} + c_1 c_2 z^{-2}][1 - (c_1 + c_2)^* z^{-1} + c_1^* c_2^* z^{-2}]$   
 $= [z + c_1 c_2 z^{-1} - (c_1 + c_2)] [z + c_1^* c_2^* z^{-1} - (c_1^* + c_2^*)] z^{-2}$

当  $c_1 = c_2^{-1}$  时,

$H_1(e^{j\omega}) = [2\cos\omega - (c_1 + c_2)][2\cos\omega - (c_1 + c_2)^*] e^{-j2\omega}$

$\angle H_1(e^{j\omega}) = -2\omega \Rightarrow H_1(z)$  可能是广义线性系统

(b)  $H_2(z)$  不存在共轭零极点  $\Rightarrow$  不能组成实系数线性差分方程

(c)  $H_3(z) = (1 - z^{-1})(1 - r_1 z^{-1})(1 - r_2 z^{-1}) \Rightarrow$  [能组成实系数线性差分方程]

$= (1 - z^{-1})[1 - (r_1 + r_2)z^{-1} + r_1 r_2 z^{-2}]$

$= (z^{\frac{1}{2}} - z^{-\frac{1}{2}})[z + r_1 r_2 z^{-1} - (r_1 + r_2)] z^{-\frac{3}{2}}$

当  $r_1 = r_2^{-1}$  时,  $H_3(e^{j\omega}) = 2\sin(\frac{\omega}{2})[2\cos\omega - (r_1 + r_2)] e^{j(-\frac{3}{2}\omega + \frac{\pi}{2})}$

$\angle H_3(e^{j\omega}) = -\frac{3}{2}\omega + \frac{\pi}{2} \Rightarrow H_3(z)$  是广义线性方程

2. (5.33) (a)  $H(z) = \frac{(1 - 2z^{-1})(1 - 0.75z^{-1})}{z^{-1}(1 - 0.5z^{-1})} = -2 \cdot (1 - 0.75z^{-1}) \cdot \frac{z^{-1} - 0.5}{z^{-1}(1 - 0.5z^{-1})}$

$H_{\min 1}(z) = -2 \cdot (1 - 0.75z^{-1})$ ,  $H_{\text{ap}}(z) = \frac{z^{-1} - 0.5}{z^{-1}(1 - 0.5z^{-1})}$

缩放尺度因子 -2,  $H_{\min 1}(z)$  和  $H_{\text{ap}}(z)$  是唯一解

因为  $H(z) = \frac{\prod_{|c_k| < 1} (1 - c_k z^{-1}) \prod_{|c_k| > 1} (1 - c_k^* z^{-1})}{\prod_{|d_k| < 1} (1 - d_k z^{-1}) \prod_{|d_k| > 1} (1 - d_k^* z^{-1})} = \underbrace{\frac{\prod_{|c_k| < 1} (1 - c_k z^{-1}) \prod_{|c_k| > 1} [1 - (c_k^*)^* z^{-1}]}{\prod_{|d_k| < 1} (1 - d_k z^{-1}) \prod_{|d_k| > 1} [1 - (d_k^*)^* z^{-1}]}}_{H_{\min 1}(z)} \cdot \underbrace{\prod_{|c_k| > 1} \frac{z^{-1} - c_k^{-1}}{1 - (c_k^*)^* z^{-1}} \prod_{|d_k| > 1} \frac{z^{-1} - d_k^*}{1 - d_k z^{-1}}}_{H_{\text{ap}}(z)} \cdot \underbrace{\prod_{|c_k| > 1} (-c_k) \prod_{|d_k| > 1} (-\frac{1}{d_k^*})}_{\text{Constant}}$

所以分解唯一

(b)  $H(z) = \frac{(1 - 2z^{-1})(1 - 0.75z^{-1})}{z^{-1}(1 - 0.5z^{-1})} = \frac{1 - 0.75z^{-1}}{(1 - 0.5z^{-1})^2} \cdot \frac{1 - 2z^{-1}}{z^{-1}} \cdot (1 - 0.5z^{-1}) = \frac{1 - 0.75z^{-1}}{(1 - 0.5z^{-1})^2} \cdot (z + z^{-1} - 2.5)$

$H_{\min 2}(z) = \frac{1 - 0.75z^{-1}}{(1 - 0.5z^{-1})^2}$ ,  $H_{\text{lp}}(z) = z + z^{-1} - 2.5$  [ $h[-1:1] = [1, -2.5, 1]$ ,  $h[n] = 0$  when  $|n| > 1$ ]

若算尺度因子,  $H_{\min 2}(z) \rightarrow 2H_{\min 2}(z)$ ,  $H_{\text{lp}}(z) \rightarrow \frac{1}{2}H_{\text{lp}}(z)$  也是另一解

分解不唯一

3. (5.43) (a)  $H(z) = \frac{z^{-2}(1 - 2z^{-1})}{2(1 - \frac{1}{2}z^{-1})} = -\frac{z^{-3}(1 - \frac{1}{2}z^{-1})}{1 - \frac{1}{2}z^{-1}}$

$|H(e^{j\omega})| = |H(z)|_{z=e^{j\omega}} = 1 \Rightarrow H(z)$  是全通系统

(b)  $H_{\min}(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} \Rightarrow h_{\min}[n] = 2^{-n}u[n]$

$H_{\max}(z) = \frac{1 - 2z^{-1}}{2} \Rightarrow h_{\max}[n] = \frac{1}{2}(\delta[n] - 2\delta[n-1])$

$H_d(z) = z^{-2} \Rightarrow h_d[n] = \delta[n-2]$

4. (5.52)  $x[n] \rightarrow X(z)$

$\alpha^n x[n] \rightarrow X(\alpha z)$

$X(\alpha z) = \frac{(1 - \frac{\alpha}{2}z^{-1})(1 - \frac{\alpha}{4}z^{-1})(1 - \frac{1}{6\alpha}z^{-1})}{1 - \frac{1}{6\alpha}z^{-1}}$   
 $= \frac{6}{5} \cdot \frac{(1 - \frac{\alpha}{2}z^{-1})(1 - \frac{\alpha}{4}z^{-1})(1 - 5\alpha z^{-1})}{1 - 6\alpha z^{-1}}$

要使  $X(\alpha z)$  是最小相位序列, 则要令

$\max(\frac{|\alpha|}{2}, \frac{|\alpha|}{4}, |5\alpha|, |6\alpha|) < 1$

$\Rightarrow |\alpha| < \frac{1}{6}$ , 结合  $\alpha^n x[n] \in \mathbb{R}$ , 得

即当  $-\frac{1}{6} < \alpha < \frac{1}{6}$  时,  $\alpha^n x[n]$  满足题意

5. (5.61)  $h_{\text{II}}[n] = h_{\text{II}}[M-n]$ ,  $0 \leq n \leq M$ ,  $M$  是奇数

$H_{\text{II}}(e^{j\omega}) = A(e^{j\omega}) e^{j(-\frac{M}{2}\omega + \theta)}$ ,  $A(e^{j\omega}) \in \mathbb{R}$

$H(z) = H_{\text{II}}(1 - z^{-1}) \Rightarrow H(e^{j\omega}) = A(e^{j\omega}) e^{j(-\frac{M}{2}\omega + \theta)} \cdot (1 - e^{-j\omega})$   
 $= A(e^{j\omega}) \cdot 2\sin\frac{\omega}{2} e^{j(-\frac{M+1}{2}\omega + \theta + \frac{\pi}{2})} \Rightarrow$  是广义线性相位系统

$H_{\text{II}}(z) \rightarrow h_{\text{II}}[n]$

$H_{\text{II}}(z)(1 - z^{-1}) \rightarrow h_{\text{II}}[n] - h_{\text{II}}[n-1] = h[n]$

当  $n < 0$  时,  $h_{\text{II}}[n] = h_{\text{II}}[n-1] = 0 \Rightarrow h[n] = 0$

当  $n > M+1$  时,  $h_{\text{II}}[n] = h_{\text{II}}[n-1] = 0 \Rightarrow h[n] = 0$

当  $0 \leq n \leq M+1$  时,  $h[M+1-n] = h[M+1-n] - h[M-n]$

$= h[n-1] - h[n] = -h[n]$  是 III 类 FIR 广义线性相位系统。